

X.

Language and Reality in Modern Physics

THROUGHOUT the history of science new discoveries and new ideas have always caused scientific disputes, have led to polemical publications criticizing the new ideas, and such criticism has often been helpful in their development; but these controversies have never before reached that degree of violence which they attained after the discovery of the theory of relativity and in a lesser degree after quantum theory. In both cases the scientific problems have finally become connected with political issues, and some scientists have taken recourse to political methods to carry their views through. This violent reaction on the recent development of modern physics can only be understood when one realizes that here the foundations of physics have started moving; and that this motion has caused the feeling that the ground would be cut from science. At the same time it probably means that one has not yet found the correct language with which to speak about the new situation and that the incorrect statements published here and there in the enthusiasm about the new discoveries have caused all kinds of misunderstanding. This is indeed a fundamental problem. The improved experimental technique of our time brings into the scope of science new as-

pects of nature which cannot be described in terms of the common concepts. But in what language, then, should they be described? The first language that emerges from the process of scientific clarification is in theoretical physics usually a mathematical language, the mathematical scheme, which allows one to predict the results of experiments. The physicist may be satisfied when he has the mathematical scheme and knows how to use it for the interpretation of the experiments. But he has to speak about his results also to nonphysicists who will not be satisfied unless some explanation is given in plain language, understandable to anybody. Even for the physicist the description in plain language will be a criterion of the degree of understanding that has been reached. To what extent is such a description at all possible? Can one speak about the atom itself? This is a problem of language as much as of physics, and therefore some remarks are necessary concerning language in general and scientific language specifically.

Language was formed during the prehistoric age among the human race as a means for communication and as a basis for thinking. We know little about the various steps in its formation; but language now contains a great number of concepts which are a suitable tool for more or less unambiguous communication about events in daily life. These concepts are acquired gradually without critical analysis by using the language, and after having used a word sufficiently often we think that we more or less know what it means. It is of course a well-known fact that the words are not so clearly defined as they seem to be at first sight and that they have only a limited range of applicability. For instance, we can speak about a piece of iron or a piece of wood, but we cannot speak about a piece of water. The word "piece" does not apply to liquid substances. Or, to mention another ex-

ample: In discussions about the limitations of concepts, Bohr likes to tell the following story: "A little boy goes into a grocer's shop with a penny in his hand and asks: 'Could I have a penny's worth of mixed sweets?' The grocer takes two sweets and hands them to the boy saying: 'Here you have two sweets. You can do the mixing yourself.'" A more serious example of the problematic relation between words and concepts is the fact that the words "red" and "green" are used even by people who are colorblind, though the ranges of applicability of these terms must be quite different for them from what they are for other people.

This intrinsic uncertainty of the meaning of words was of course recognized very early and has brought about the need for definitions, or—as the word "definition" says—for the setting of boundaries that determine where the word is to be used and where not. But definitions can be given only with the help of other concepts, and so one will finally have to rely on some concepts that are taken as they are, unanalyzed and undefined.

In Greek philosophy the problem of the concepts in language has been a major theme since Socrates, whose life was—if we can follow Plato's artistic representation in his dialogues—a continuous discussion about the content of the concepts in language and about the limitations in modes of expression. In order to obtain a solid basis for scientific thinking, Aristotle in his logic started to analyze the forms of language, the formal structure of conclusions and deductions independent of their content. In this way he reached a degree of abstraction and precision that had been unknown up to that time in Greek philosophy and he thereby contributed immensely to the clarification, to the establishment of order in our methods of thought. He actually created the basis for the scientific language.

On the other hand, this logical analysis of language again involves the danger of an oversimplification. In logic the attention is drawn to very special structures, unambiguous connections between premises and deductions, simple patterns of reasoning, and all the other structures of language are neglected. These other structures may arise from associations between certain meanings of words; for instance, a secondary meaning of a word which passes only vaguely through the mind when the word is heard may contribute essentially to the content of a sentence. The fact that every word may cause many only half-conscious movements in our mind can be used to represent some part of reality in the language much more clearly than by the use of the logical patterns. Therefore, the poets have often objected to this emphasis in language and in thinking on the logical pattern, which—if I interpret their opinions correctly—can make language less suitable for its purpose. We may recall for instance the words in Goethe's *Faust* which Mephistopheles speaks to the young student (quoted from the translation by Anna Swanwick):

Waste not your time, so fast it flies;
Method will teach you time to win;
Hence, my young friend, I would advise,
With college logic to begin.
Then will your mind be so well brac'd,
In Spanish boots so tightly lac'd,
That on 'twill circumspectly creep,
Thought's beaten track securely keep,
Nor will it, ignis-fatuus like,
Into the path of error strike.
Then many a day they'll teach you how
The mind's spontaneous acts, till now

As eating and as drinking free,
Require a process;—one, two, three!
In truth the subtle web of thought
Is like the weaver's fabric wrought,
One treadle moves a thousand lines,
Swift dart the shuttles to and fro,
Unseen the threads unnumber'd flow,
A thousand knots one stroke combines.
Then forward steps your sage to show,
And prove to you it must be so;
The first being so, and so the second.
The third and fourth deduc'd we see;
And if there were no first and second,
Nor third nor fourth would ever be.
This, scholars of all countries prize,
Yet 'mong themselves no weavers rise.
Who would describe and study aught alive,
Seeks first the living spirit thence to drive:
Then are the lifeless fragments in his hand,
There only fails, alas!—the spirit-band.

This passage contains an admirable description of the structure of language and of the narrowness of the simple logical patterns.

On the other hand, science must be based upon language as the only means of communication and there, where the problem of unambiguity is of greatest importance, the logical patterns must play their role. The characteristic difficulty at this point may be described in the following way. In natural science we try to derive the particular from the general, to understand the particular phenomenon as caused by simple general laws. The general laws when formulated in the language can contain only a few simple concepts—else the law would not be simple and general. From these concepts are derived an infinite variety of

possible phenomena, not only qualitatively but with complete precision with respect to every detail. It is obvious that the concepts of ordinary language, inaccurate and only vaguely defined as they are, could never allow such derivations. When a chain of conclusions follows from given premises, the number of possible links in the chain depends on the precision of the premises. Therefore, the concepts of the general laws must in natural science be defined with complete precision, and this can be achieved only by means of mathematical abstraction.

In other sciences the situation may be somewhat similar in so far as rather precise definitions are also required; for instance, in law. But here the number of links in the chain of conclusions need not be very great, complete precision is not needed, and rather precise definitions in terms of ordinary language are sufficient.

In theoretical physics we try to understand groups of phenomena by introducing mathematical symbols that can be correlated with facts, namely, with the results of measurements. For the symbols we use names that visualize their correlation with the measurement. Thus the symbols are attached to the language. Then the symbols are interconnected by a rigorous system of definitions and axioms, and finally the natural laws are expressed as equations between the symbols. The infinite variety of solutions of these equations then corresponds to the infinite variety of particular phenomena that are possible in this part of nature. In this way the mathematical scheme represents the group of phenomena so far as the correlation between the symbols and the measurements goes. It is this correlation which permits the expression of natural laws in the terms of common language, since our experiments consisting of actions and observations can always be described in ordinary language.

Still, in the process of expansion of scientific knowledge the language also expands; new terms are introduced and the old ones are applied in a wider field or differently from ordinary language. Terms such as "energy," "electricity," "entropy" are obvious examples. In this way we develop a scientific language which may be called a natural extension of ordinary language adapted to the added fields of scientific knowledge.

During the past century a number of new concepts have been introduced in physics, and in some cases it has taken considerable time before the scientists have really grown accustomed to their use. The term "electromagnetic field," for instance, which was to some extent already present in Faraday's work and which later formed the basis of Maxwell's theory, was not easily accepted by the physicists, who directed their attention primarily to the mechanical motion of matter. The introduction of the concept really involved a change in scientific ideas as well, and such changes are not easily accomplished.

Still, all the concepts introduced up to the end of the last century formed a perfectly consistent set applicable to a wide field of experience, and, together with the former concepts, formed a language which not only the scientists but also the technicians and engineers could successfully apply in their work. To the underlying fundamental ideas of this language belonged the assumptions that the order of events in time is entirely independent of their order in space, that Euclidean geometry is valid in real space, and that the events "happen" in space and time independently of whether they are observed or not. It was not denied that every observation had some influence on the phenomenon to be observed but it was generally assumed that by doing the experiments cautiously this influence could be made arbitrarily small. This seemed in fact a necessary condition for

the ideal of objectivity which was considered as the basis of all natural science.

Into this rather peaceful state of physics broke quantum theory and the theory of special relativity as a sudden, at first slow and then gradually increasing, movement in the foundations of natural science. The first violent discussions developed around the problems of space and time raised by the theory of relativity. How should one speak about the new situation? Should one consider the Lorentz contraction of moving bodies as a real contraction or only as an apparent contraction? Should one say that the structure of space and time was really different from what it had been assumed to be or should one only say that the experimental results could be connected mathematically in a way corresponding to this new structure, while space and time, being the universal and necessary mode in which things appear to us, remain what they had always been? The real problem behind these many controversies was the fact that no language existed in which one could speak consistently about the new situation. The ordinary language was based upon the old concepts of space and time and this language offered the only unambiguous means of communication about the setting up and the results of the measurements. Yet the experiments showed that the old concepts could not be applied everywhere.

The obvious starting point for the interpretation of the theory of relativity was therefore the fact that in the limiting case of small velocities (small compared with the velocity of light) the new theory was practically identical with the old one. Therefore, in this part of the theory it was obvious in which way the mathematical symbols had to be correlated with the measurements and with the terms of ordinary language; actually it was only through this correlation that the Lorentz transformation had

been found. There was no ambiguity about the meaning of the words and the symbols in this region. In fact this correlation was already sufficient for the application of the theory to the whole field of experimental research connected with the problem of relativity. Therefore, the controversial questions about the "real" or the "apparent" Lorentz contraction, or about the definition of the word "simultaneous" etc., did not concern the facts but rather the language.

With regard to the language, on the other hand, one has gradually recognized that one should perhaps not insist too much on certain principles. It is always difficult to find general convincing criteria for which terms should be used in the language and how they should be used. One should simply wait for the development of the language, which adjusts itself after some time to the new situation. Actually in the theory of special relativity this adjustment has already taken place to a large extent during the past fifty years. The distinction between "real" and "apparent" contraction, for instance, has simply disappeared. The word "simultaneous" is used in line with the definition given by Einstein, while for the wider definition discussed in an earlier chapter the term "at a space-like distance" is commonly used, etc.

In the theory of general relativity the idea of a non-Euclidean geometry in real space was strongly contradicted by some philosophers who pointed out that our whole method of setting up the experiments already presupposed Euclidean geometry.

In fact if a mechanic tries to prepare a perfectly plane surface, he can do it in the following way. He first prepares three surfaces of, roughly, the same size which are, roughly, plane. Then he tries to bring any two of the three surfaces into contact by putting them against each other in different relative positions. The

degree to which this contact is possible on the whole surface is a measure of the degree of accuracy with which the surfaces can be called "plane." He will be satisfied with his three surfaces only if the contact between any two of them is complete everywhere. If this happens one can prove mathematically that Euclidean geometry holds on the three surfaces. In this way, it was argued, Euclidean geometry is just *made* correct by our own measures.

From the point of view of general relativity, of course, one can answer that this argument proves the validity of Euclidean geometry only in small dimensions, in the dimensions of our experimental equipment. The accuracy with which it holds in this region is so high that the above process for getting plane surfaces can always be carried out. The extremely slight deviations from Euclidean geometry which still exist in this region will not be realized since the surfaces are made of material which is not strictly rigid but allows for very small deformations and since the concept of "contact" cannot be defined with complete precision. For surfaces on a cosmic scale the process that has been described would just not work; but this is not a problem of experimental physics.

Again, the obvious starting point for the physical interpretation of the mathematical scheme in general relativity is the fact that the geometry is very nearly Euclidean in small dimensions; the theory approaches the classical theory in this region. Therefore, here the correlation between the mathematical symbols and the measurements and the concepts in ordinary language is unambiguous. Still, one can speak about a non-Euclidean geometry in large dimensions. In fact a long time before the theory of general relativity had even been developed the possibility of a non-Euclidean geometry in real space seems to have been con-

sidered by the mathematicians, especially by Gauss in Göttingen. When he carried out very accurate geodetic measurements on a triangle formed by three mountains—the Brocken in the Harz Mountains, the Inselberg in Thuringia, and the Hohenhagen near Göttingen—he is said to have checked very carefully whether the sum of the three angles was actually equal to 180 degrees; and that he considered a difference which would prove deviations from Euclidean geometry as being possible. Actually he did not find any deviations within his accuracy of measurement.

In the theory of general relativity the language by which we describe the general laws actually now follows the scientific language of the mathematicians, and for the description of the experiments themselves we can use the ordinary concepts, since Euclidean geometry is valid with sufficient accuracy in small dimensions.

The most difficult problem, however, concerning the use of the language arises in quantum theory. Here we have at first no simple guide for correlating the mathematical symbols with concepts of ordinary language; and the only thing we know from the start is the fact that our common concepts cannot be applied to the structure of the atoms. Again the obvious starting point for the physical interpretation of the formalism seems to be the fact that the mathematical scheme of quantum mechanics approaches that of classical mechanics in dimensions which are large as compared to the size of the atoms. But even this statement must be made with some reservations. Even in large dimensions there are many solutions of the quantum-theoretical equations to which no analogous solutions can be found in classical physics. In these solutions the phenomenon of the “interference of probabilities” would show up, as was discussed in the earlier chapters;

it does not exist in classical physics. Therefore, even in the limit of large dimensions the correlation between the mathematical symbols, the measurements, and the ordinary concepts is by no means trivial. In order to get to such an unambiguous correlation one must take another feature of the problem into account. It must be observed that the system which is treated by the methods of quantum mechanics is in fact a part of a much bigger system (eventually the whole world); it is interacting with this bigger system; and one must add that the microscopic properties of the bigger system are (at least to a large extent) unknown. This statement is undoubtedly a correct description of the actual situation. Since the system could not be the object of measurements and of theoretical investigations, it would in fact not belong to the world of phenomena if it had no interactions with such a bigger system of which the observer is a part. The interaction with the bigger system with its undefined microscopic properties then introduces a new statistical element into the description—both the quantum-theoretical and the classical one—of the system under consideration. In the limiting case of the large dimensions this statistical element destroys the effects of the “interference of probabilities” in such a manner that now the quantum-mechanical scheme really approaches the classical one in the limit. Therefore, at this point the correlation between the mathematical symbols of quantum theory and the concepts of ordinary language is unambiguous, and this correlation suffices for the interpretation of the experiments. The remaining problems again concern the language rather than the facts, since it belongs to the concept “fact” that it can be described in ordinary language.

But the problems of language here are really serious. We wish to speak in some way about the structure of the atoms and not

only about the "facts"—the latter being, for instance, the black spots on a photographic plate or the water droplets in a cloud chamber. But we cannot speak about the atoms in ordinary language.

The analysis can now be carried further in two entirely different ways. We can either ask which language concerning the atoms has actually developed among the physicists in the thirty years that have elapsed since the formulation of quantum mechanics. Or we can describe the attempts for defining a precise scientific language that corresponds to the mathematical scheme.

In answer to the first question one may say that the concept of complementarity introduced by Bohr into the interpretation of quantum theory has encouraged the physicists to use an ambiguous rather than an unambiguous language, to use the classical concepts in a somewhat vague manner in conformity with the principle of uncertainty, to apply alternatively different classical concepts which would lead to contradictions if used simultaneously. In this way one speaks about electronic orbits, about matter waves and charge density, about energy and momentum, etc., always conscious of the fact that these concepts have only a very limited range of applicability. When this vague and unsystematic use of the language leads into difficulties, the physicist has to withdraw into the mathematical scheme and its unambiguous correlation with the experimental facts.

This use of the language is in many ways quite satisfactory, since it reminds us of a similar use of the language in daily life or in poetry. We realize that the situation of complementarity is not confined to the atomic world alone; we meet it when we reflect about a decision and the motives for our decision or when we have the choice between enjoying music and analyzing its structure. On the other hand, when the classical concepts are

used in this manner, they always retain a certain vagueness, they acquire in their relation to "reality" only the same statistical significance as the concepts of classical thermodynamics in its statistical interpretation. Therefore, a short discussion of these statistical concepts of thermodynamics may be useful.

The concept "temperature" in classical thermodynamics seems to describe an objective feature of reality, an objective property of matter. In daily life it is quite easy to define with the help of a thermometer what we mean by stating that a piece of matter has a certain temperature. But when we try to define what the temperature of an atom could mean we are, even in classical physics, in a much more difficult position. Actually we cannot correlate this concept "temperature of the atom" with a well-defined property of the atom but have to connect it at least partly with our insufficient knowledge of it. We can correlate the value of the temperature with certain statistical expectations about the properties of the atom, but it seems rather doubtful whether an expectation should be called objective. The concept "temperature of the atom" is not much better defined than the concept "mixing" in the story about the boy who bought mixed sweets.

In a similar way in quantum theory all the classical concepts are, when applied to the atom, just as well and just as little defined as the "temperature of the atom"; they are correlated with statistical expectations; only in rare cases may the expectation become the equivalent of certainty. Again, as in classical thermodynamics, it is difficult to call the expectation objective. One might perhaps call it an objective tendency or possibility, a "potentia" in the sense of Aristotelian philosophy. In fact, I believe that the language actually used by physicists when they speak about atomic events produces in their minds similar

notions as the concept "potentia." So the physicists have gradually become accustomed to considering the electronic orbits, etc., not as reality but rather as a kind of "potentia." The language has already adjusted itself, at least to some extent, to this true situation. But it is not a precise language in which one could use the normal logical patterns; it is a language that produces pictures in our mind, but together with them the notion that the pictures have only a vague connection with reality, that they represent only a tendency toward reality.

The vagueness of this language in use among the physicists has therefore led to attempts to define a different precise language which follows definite logical patterns in complete conformity with the mathematical scheme of quantum theory. The result of these attempts by Birkhoff and Neumann and more recently by Weizsäcker can be stated by saying that the mathematical scheme of quantum theory can be interpreted as an extension or modification of classical logic. It is especially one fundamental principle of classical logic which seems to require a modification. In classical logic it is assumed that, if a statement has any meaning at all, either the statement or the negation of the statement must be correct. Of "here is a table" or "here is not a table," either the first or the second statement must be correct. "Tertium non datur," a third possibility does not exist. It may be that we do not know whether the statement or its negation is correct; but in "reality" one of the two is correct.

In quantum theory this law "tertium non datur" is to be modified. Against any modification of this fundamental principle one can of course at once argue that the principle is assumed in common language and that we have to speak at least about our eventual modification of logic in the natural language. Therefore, it would be a self-contradiction to describe in natural lan-

guage a logical scheme that does not apply to natural language. There, however, Weizsäcker points out that one may distinguish various levels of language.

One level refers to the objects—for instance, to the atoms or the electrons. A second level refers to statements about objects. A third level may refer to statements about statements about objects, etc. It would then be possible to have different logical patterns at the different levels. It is true that finally we have to go back to the natural language and thereby to the classical logical patterns. But Weizsäcker suggests that classical logic may be in a similar manner a priori to quantum logic, as classical physics is to quantum theory. Classical logic would then be contained as a kind of limiting case in quantum logic, but the latter would constitute the more general logical pattern.

The possible modification of the classical logical pattern shall, then, first refer to the level concerning the objects. Let us consider an atom moving in a closed box which is divided by a wall into two equal parts. The wall may have a very small hole so that the atom can go through. Then the atom can, according to classical logic, be either in the left half of the box or in the right half. There is no third possibility: “tertium non datur.” In quantum theory, however, we have to admit—if we use the words “atom” and “box” at all—that there are other possibilities which are in a strange way mixtures of the two former possibilities. This is necessary for explaining the results of our experiments. We could, for instance, observe light that has been scattered by the atom. We could perform three experiments: first the atom is (for instance, by closing the hole in the wall) confined to the left half of the box, and the intensity distribution of the scattered light is measured; then it is confined to the right half and again the scattered light is measured; and finally the

atom can move freely in the whole box and again the intensity distribution of the scattered light is measured. If the atom would always be in either the left half or the right half of the box, the final intensity distribution should be a mixture (according to the fraction of time spent by the atom in each of the two parts) of the two former intensity distributions. But this is in general not true experimentally. The real intensity distribution is modified by the "interference of probabilities"; this has been discussed before.

In order to cope with this situation Weizsäcker has introduced the concept "degree of truth." For any simple statement in an alternative like "The atom is in the left (or in the right) half of the box" a complex number is defined as a measure for its "degree of truth." If the number is 1, it means that the statement is true; if the number is 0, it means that it is false. But other values are possible. The absolute square of the complex number gives the probability for the statement's being true; the sum of the two probabilities referring to the two parts in the alternative (either "left" or "right" in our case) must be unity. But each pair of complex numbers referring to the two parts of the alternative represents, according to Weizsäcker's definitions, a "statement" which is certainly true if the numbers have just these values; the two numbers, for instance, are sufficient for determining the intensity distribution of scattered light in our experiment. If one allows the use of the term "statement" in this way one can introduce the term "complementarity" by the following definition: Each statement that is not identical with either of the two alternative statements—in our case with the statements: "the atom is in the left half" or "the atom is in the right half of the box"—is called complementary to these statements. For each complementary statement the question whether

the atom is left or right is not decided. But the term "not decided" is by no means equivalent to the term "not known." "Not known" would mean that the atom is "really" left or right, only we do not know where it is. But "not decided" indicates a different situation, expressible only by a complementary statement.

This general logical pattern, the details of which cannot be described here, corresponds precisely to the mathematical formalism of quantum theory. It forms the basis of a precise language that can be used to describe the structure of the atom. But the application of such a language raises a number of difficult problems of which we shall discuss only two here: the relation between the different "levels" of language and the consequences for the underlying ontology.

In classical logic the relation between the different levels of language is a one-to-one correspondence. The two statements, "The atom is in the left half" and "It is true that the atom is in the left half," belong logically to different levels. In classical logic these statements are completely equivalent, i.e., they are either both true or both false. It is not possible that the one is true and the other false. But in the logical pattern of complementarity this relation is more complicated. The correctness or incorrectness of the first statement still implies the correctness or incorrectness of the second statement. But the incorrectness of the second statement does not imply the incorrectness of the first statement. If the second statement is incorrect, it may be undecided whether the atom is in the left half; the atom need not necessarily be in the right half. There is still complete equivalence between the two levels of language with respect to the correctness of a statement, but not with respect to the incorrectness. From this connection one can understand the persistence of the classical laws in quantum theory: wherever a definite result can be derived

in a given experiment by the application of the classical laws the result will also follow from quantum theory, and it will hold experimentally.

The final aim of Weizsäcker's attempt is to apply the modified logical patterns also in the higher levels of language, but these questions cannot be discussed here.

The other problem concerns the ontology that underlies the modified logical patterns. If the pair of complex numbers represents a "statement" in the sense just described, there should exist a "state" or a "situation" in nature in which the statement is correct. We will use the word "state" in this connection. The "states" corresponding to complementary statements are then called "coexistent states" by Weizsäcker. This term "coexistent" describes the situation correctly; it would in fact be difficult to call them "different states," since every state contains to some extent also the other "coexistent states." This concept of "state" would then form a first definition concerning the ontology of quantum theory. One sees at once that this use of the word "state," especially the term "coexistent state," is so different from the usual materialistic ontology that one may doubt whether one is using a convenient terminology. On the other hand, if one considers the word "state" as describing some potentiality rather than a reality—one may even simply replace the term "state" by the term "potentiality"—then the concept of "coexistent potentialities" is quite plausible, since one potentiality may involve or overlap other potentialities.

All these difficult definitions and distinctions can be avoided if one confines the language to the description of facts, i.e., experimental results. However, if one wishes to speak about the atomic particles themselves one must either use the mathematical scheme as the only supplement to natural language or one must

combine it with a language that makes use of a modified logic or of no well-defined logic at all. In the experiments about atomic events we have to do with things and facts, with phenomena that are just as real as any phenomena in daily life. But the atoms or the elementary particles themselves are not as real; they form a world of potentialities or possibilities rather than one of things or facts.